A REVIEW OF FAST REACTOR DEVELOPMENT IN THE EURATOM COMMUNITY

A. DE STORDEUR*

1. Introduction

Interest in fast reactors in the Community dates back some 10 years when the French started their first conceptual design study of the Rapsodie reactor and the Belgians sent a team of engineers and physicists to contribute to the design of the Fermi reactor in Detroit. Since then, Germany started a very active programme in 1959–60 in its nuclear research centre of Karlsruhe. Italy started its efforts in 1962–3 and the Netherlands in 1964–5.

The Euratom Commission included the study of fast plutonium reactors in its official research programme in 1959 and proposed a policy of association with the programmes which had started on a national scale: they would be converted into broader based enterprises in which the national organizations and Euratom would undertake together the financing, management, and maintenance of the joint working teams. This policy was ultimately approved by the Community's Council of Ministers when it decided upon Euratom's second 5-year plan (1963–7) and association contracts were successively signed with the French Commissariat à l'Energie Atomique (CEA), the German Gesellschaft für Kernforschung (GfK) in Karlsruhe, and the Italian Comitato Nationale per l'Energia Nucleare (CNEN).

After Euratom's second 5-year plan was revised by the Council of Ministers in mid-1965, it became possible to enlarge the associations to include all five active nuclear Member States of the Community and to conclude new contracts with the Belgian Commissariat à l'Energie Atomique and with a Dutch grouping of the governmental organizations Toegepast Natuurwetenschappelijke Onderzook (TNO) and Reactor Centrum Nederland (RCN); these associations in turn subcontracted work to Belgo Nucléaire and the Belgian Centre d'Etude Nucléaire of Mol and to Neratoom, respectively.

The Commission devotes to its associations practically the total funds and

* Euratom—Communauté Européenne de l'Energie Atomique.

manpower which the Council of Ministers of the Six have appropriated for its fast reactor effort during its second 5-year period; it has no project of its own but helps the actions of the associations in different important areas by its common research centre, and mainly by the Geel (BCMN), Ispra, and Transuranium Institute establishments.

Considering that the 82·5 million EMA units of account,* of the Euratom second 5-year programme, represent only about one-third of the overall Community funds, since the five associates of the Commission contribute with their own resources about two-thirds of the effort, it can be said that the overall effort of the Community between 1963 and 1967 will be of the order of $50 million per year; this figure indicates the importance attached to the fast breeder reactor's development.

This importance is justified by the promising conclusions drawn from the economic evaluations performed in the Community on designs of large size power reactors evolved by Cadarache and Karlsruhe, which confirm those obtained in the U.S.A. and the U.K.; it is also justified by the results obtained from the nuclear resources strategy studies, indicating the ultimate need for fast breeder power reactors in the ever expanding power programme in the Community: studies performed by the Euratom Commission for its indicative programme of power demand,[1] that of Karlsruhe published at the last Foratom meeting[2] as well as those performed jointly by the French Commissariat à l'Energie Atomique and Electricité de France, and by Belgo Nucléaire.[3]

The overall objective of the different programmes is to achieve competitive fast breeder reactor power by the late 1970s. It is thought that stations of the order of 1000 MW(e) will be installed at that time and therefore this nominal size has been selected to guide the development efforts; construction of prototype reactors in the range of 100–300 MW(e) could be envisaged to start in 1969 approximately.

The purpose of the present paper is to review the programmes being carried out or envisaged in this field in the Community and to serve as an introduction to the more specific papers presented in this Conference by several Community contributors;[4, 5, 6, 7, 8, 9] the different areas of the development will be reviewed, i.e. large reactor and core design, physics and safety, reactor system development, and fuel preparation and reprocessing.

2. Large Reactor and Core Design

The two main Community centres of activity in this area are Cadarache and Karlsruhe which have evolved preliminary designs of 1000 MW(e) power

* The value of one unit of account (u.a.) is that of 0·888 g of refined gold, and approximately $1.

stations and are presently engaged in the design of prototype reactors whose construction could possibly start in 1969.

Whereas Karlsruhe has been successively evaluating designs based on gas, sodium, and dry steam cooling and is at present concentrating with the help of industry and the Euratom–Belgian association on the two latter ones,[5, 10, 11] Cadarache has from the start been concentrating on the sodium cooling line[6] the first step being the 20 MW(th) Rapsodie reactor which will go critical in Cadarache at the end of this year. Both projects are based on the use of plutonium–uranium oxide, vented-to-plenum fuels; the adaption of these designs to the vented-to-coolant type of fuel is being studied at Bologna under the Euratom–CNEN association.

Different variants have been or are still being considered for both types of coolant now selected. For sodium cooling studies were made on pancaking of the core, multizone fuelling and fuel management, tank or external loop concepts, different schemes of fuel handling equipment, etc.; at the present time the designers favour core shapes with height to diameter ratios of about 0·35, multizone core systems and external loop cooling.

The steam reactor designs studied at Karlsruhe[11] and Belgo Nucléaire[12] are based on the use of Loeffler cycles; both pressure vessel and modular pressure tubes have been considered. The present design[11] gives preference to the pressure vessel with underwater loading and unloading.

These studies have indicated that both sodium and steam designs are basically feasible and safe, and their economic evaluation has indicated that they would be competitive with conventional or thermal reactor power stations built in the 1980s.

These programmes draw heavily from the experience acquired on the design and building of the Rapsodie reactor as well as from the KNK[13] and HDR[14] reactors now being built by the Federal Government of Germany in the framework of its thermal reactor development programme.

3. Physics and Safety

As is the case outside the Community, reactor physics and safety evaluation have received considerable attention, as can be seen from listing the facilities now available in the Community.

STARK (Schnell–Thermischer–Argonaut Reaktor Karlsruhe)[15] which went critical in July 1964, after transformation of an existing Argonaut, essentially by replacing the central graphite column by a fast neutron zone. The first series of experiments has been completed on a natural uranium central zone and the second series is presently under way with an enriched uranium central loading. This experiment has produced such original results on the noise analysis method[16] that its operation will be continued in parallel with that of the critical experiment SNEAK and will also be used for the mixed spectrum steam cooled reactor study.

SUAK (Schnelle Unterkritische Anordnung Karlsruhe)[17] is a pulsed fast neutron subcritical assembly which started operation in mid-1964 and techniques developed at the Karlsruhe Centre are being extensively used on it: subcritical core configuration with enrichment of about 10% is being pulsed with 14 MeV neutrons from a *d*, *t* target reaction, permitting neutron spectra detection in the keV range.

Harmonie[18] is a 2 kW fast source reactor which went critical in August 1965 in Cadarache and is being used for instrument calibration as well as for driving fast subcritical assemblies or for studying shielding. This facility which is basically similar to the Argonne AFBR reactor is equipped with a unique feature permitting the extraction of the core from the shielding, thus allowing for unshielded type experiments of the Los Alamos Hydro type to be carried out.

SNEAK (Schnelle Null Energie Anordnung Karlsruhe)[19] is a large, plutonium fuelled critical assembly which is in the last stages of the approach to criticality on an enriched uranium loading.

It is a fixed vertical machine using suspended tubes (2 × 2 in. section) in which square platelets of the Zebra type are stacked. Plutonium plates are made of UO_2–PuO_2 flat pellets contained in a square box. The maximum core volume is about 2000 litres with an 80 cm thick blanket. Provision has been made for installing a central heated loop for Doppler measurements.

Masurca (Maquette Surgenerateur Cadarache)[20] is the Cadarache large, plutonium fuelled critical assembly whose construction is nearing completion and whose criticality on a plutonium loading is now scheduled for the end of 1966. It is also a fixed vertical machine using suspended tubes (4×4 in. section) but its loading will be made up of square rods (and cylindrical rods for plutonium elements made out of uranium–plutonium–iron alloy) stacked vertically end to end in the tubes. Maximum core volume is about 5000 litres with an axial blanket of 60 cm and a radial blanket of 50 cm. Provision has also been made for installing a central heated loop for Doppler measurements.

In addition to these facilities, the Euratom–GfK association participates in the joint American and European venture of the SEFOR reactor,[21] which should go critical in Fayetteville (Arkansas) by autumn 1967. Its experiments on Doppler measurements at high temperature with temperature distribution and spectra representative of large fast reactors should be finished by the end of 1970 and will contribute greatly to the understanding of this important safety aspect.

These experimental facilities are supported by important teams of experimental and theoretical physicists in Bologna, Brussels, Cadarache, Karlsruhe, and Saclay, making full use of the computing equipment they have available (IBM 7094, IBM 1620 and Ferranti Mercury, IBM 704, IBM 707, and IBM

7090, respectively), and applying the data to the design of large fast reactors.

For the determination of basic data on cross-sections, many centres of the Community are involved in the tasks distributed by the European–American Nuclear Data Committee: Karlsruhe, Cadarache, Saclay, Bologna, and Padova and the Central Bureau of Nuclear Measurement of Euratom in Geel.

The evaluation of fast reactor safety in the Community has been so far mainly confined to the theoretical approach of the system analysis involving the dynamics of the reactors being designed. Experimental work has been started in the field of sodium boiling in the Euratom–Ispra research establishment,(22) Karlsruhe, Cadarache, and Bologna together with the CNEN establishment of Casaccia.

4. Reactor System Development

This section briefly covers the reactor material as well as the component development needed for large fast power reactors. As already mentioned, information is gained from existing experience in the Community, from the Rapsodie programme in association with Euratom, and from material developed for the KNK and HDR programmes under exclusive German Government funding.

The materials being considered and developed for the sodium reactor system include austenitic as well as ferritic steels (AISI 316 stainless steel utilized in Rapsodie, stabilized 2·25 Cr, 1 Mo, 1 Nb, 0·5 Ni ferritic steel in KNK). Other materials include principally different SS and some Ni alloys: German type 4988 and 4961 stainless steels, Inconel 625, Hastalloy and 80–20 Ni–Cr base alloys.

Corrosion and mass transfer experiments are being carried out in the Euratom associations at Saclay and Fontenay-aux-Roses, at Karlsruhe, and at the TNO laboratories in Delft and Apeldoorn. They make use of a wealth of relatively small size loops already existing or under construction, working in either isothermal or non-isothermal conditions (one loop in Saclay, four in Grenoble, three in Karlsruhe, and eight in TNO).

Component development in the Community associations is being carried out at Cadarache,(9) Karlsruhe, TNO, and Bologna in direct liaison with industrial organizations.

The experimental equipment available in the associations includes the Rapsodie mock-ups of Cadarache (1 and 10 MW(th) loops, core vessel mock-up, and thermal shock loops) the 5 MW(th) steam generating facility of Grand-Quevilly, the 3 MW(th) Loeffler cycle facility of Karlsruhe, and the thermal shock facility of TNO. Larger facilities for testing heated sodium steam generating equipment are presently being considered by the Bologna and TNO associations and their design has started.

TABLE 1. CHARACTERISTICS OF UO_2-PuO_2 FUELS DEVELOPED IN THE COMMUNITY

	Euratom–CEA association		Euratom–GfK association (sodium cooling 1000 MW(e))	Euratom–CNEN association	Euratom–Belgium association
	Rapsodie	1000 MW(e) studies			
Outside pin diameter (mm)	6·7	6	6·7	6·5	6·35
Clad thickness (mm)	0·45	0·4	0·35	0·3	0·38
Clad material	316 L SS	316 L SS	Incoloy 800	316 L SS	347 SS
Oxide density (% theoretical)	96	80–85	90	85	85–90
Fabrication method	Pellets	Pellets	Pellets or vibrocompacted	Vibrocompacted	Pellets
Active zone length (mm)	340	700	955	800	775
Upper blanket length (mm)	270	Approx. 500	400	450	—
Lower blanket length (mm)	270	Approx. 500	400	450	254
Position of fission gas plenum	Top	Top	Bottom	Vented to coolant	Top
Length of plenum (mm)	145	500	800	—	200

5. Fuel Preparation and Reprocessing

The main fuel development activities of the Community associations is concentrated on the uranium plutonium oxide, vented-to-plenum fuels. It involves the specialized Fontenay-aux-Roses and Cadarache Laboratories of CEA, the Karlsruhe Centre assisted by the Euratom Transuranium Institute, and the CEN–Belgo Nucléaire plutonium Laboratory in Mol. The target is the development of a high burnup fuel capable of withstanding irradiations of about 100,000 MWd/t (as in the U.S.A. and the U.K.).

Cadarache and Karlsruhe are working along the lines of their reference design reactors for either sodium or steam cooling; Mol's work is directed toward the development of a plutonium core for the Fermi reactor as a contribution to a multi-national programme organized by APDA. The main characteristics of the fuel being developed by the three associations are given in Table 1. For the Cadarache association this work is the extension of the Rapsodie oxide fuel development programme in which burnups of the order of 40,000 MWd/t have been obtained. Some important results have been recently obtained at Karlsruhe as will be reported at the meeting by Dr. Kummerer.(7)

So far, the experimental studies have used thermal reactors such as the Saclay EL-3, the Karlsruhe FR-2 or the Mol BR-2. Recently, the central hole of BR-2 has been equipped with a thermal neutron shielded sodium loop which will also be used for oxide irradiations.

Also, the irradiation possibilities of the Fermi reactor contracted for Euratom are to be extensively used and more than 100 fuel pin specimens are being prepared at present for insertion at the end of the year. Eventually, Rapsodie will also be used for experimental fuel testing, in 1968 when its start-up programme will have ended. Karlsruhe is also envisaging the possibility of equipping the KNK reactor with a mixed spectrum core and to irradiate specimens with fast neutrons in 1970–1.

This effort on vented-to-plenum oxide fuels is complemented by the work of the Bologna association on the vented-to-coolant oxide fuels which started about 1 year ago; the development programme currently involves the irradiation of encapsulated pins in thermal reactors (Avogadro in Saluggia and Br-2) and in the Fermi reactor. This programme is to be followed by in-pile loop irradiation, for which a preliminary design of a specialized reactor (PEC—Prova de elementi de combustible) was evolved and will be considered for possible construction in the very near future.

It should be mentioned that carbide and nitride fuels are being studied at Fontenay which is carrying out an irradiation programme in the fast neutron loop of BR-2 on mixed carbide–nitride systems; basic studies on these elements are starting at the Euratom Transuranium Institute.

Fast reactor fuel reprocessing studies in the Euratom associations are, at present, limited to the Karlsruhe efforts in the development of a relatively short-cycle (100 days cooling) aqueous reprocessing method, based on either the classical TBP or the amine solvents, and to the Mol work on the fluoride volatility process applied to high plutonium contents which complements their several years' work in this field for thermal reactor fuels; for Rapsodie it has been decided to reprocess the fuel by a classical aqueous method for which CEA built a special head end unit in its centre of Cap de la Hague.

6. Summary

Table 2 presents in a condensed form the preceding chapters on the areas of work being carried out by the five associations in which Euratom is involved.

TABLE 2. SUMMARY OF THE COMMUNITY FAST REACTORS EFFORT

	CEA–Eur association	GfK–Eur association	CNEN–Eur association	TNO/RCN–Eur association	Belgium–Eur association
Reactor and core design	×	×	×		×
Experimental reactor physics	×	×			
Theoretical reactor physics	×	×	×		×
Material development	×	×		×	
Components development	×	×	×	×	
Fuel development	×	×	×		×
Reprocessing		×			×

It indicates the importance of the effort devoted by the Community (approximately $50 million per year between 1963 and 1967) to fast reactors.

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